

SIPMT Playbook

Complete application help, workflows, KPI definitions, and operational guidance.

What this document contains

- Detailed guidance for every application function exposed in the Streamlit interface.
- Focused screenshots captured from the current application state.
- KPI and chart definitions, operational notes, configuration guidance, and troubleshooting tips.

Playbook purpose

This document is intended as the printable operating guide for SIPMT users, analysts, and administrators.

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Home - Overview Dashboard

Chapter Focus

The Home page is your workspace dashboard, providing at-a-glance metrics and overview of your data. Metrics Overview Document Scope Map

templates automatically, and visualise locations on an interactive map.

Documents
49

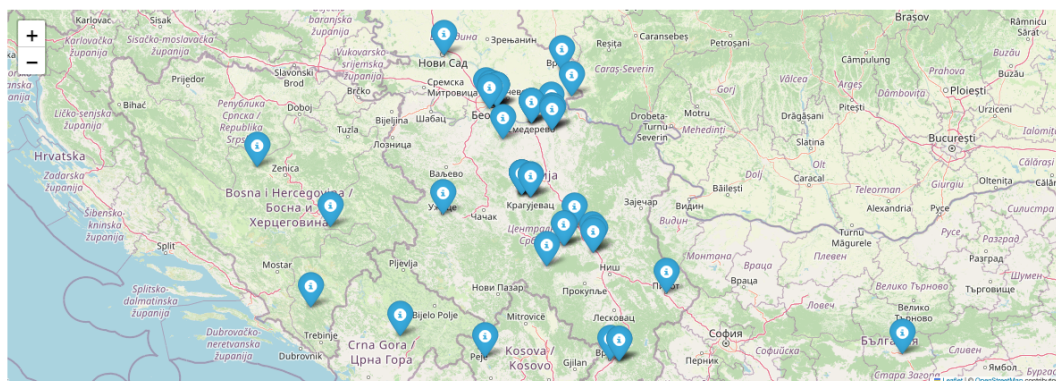
Templates
0

Reports
0

Changes detected
0

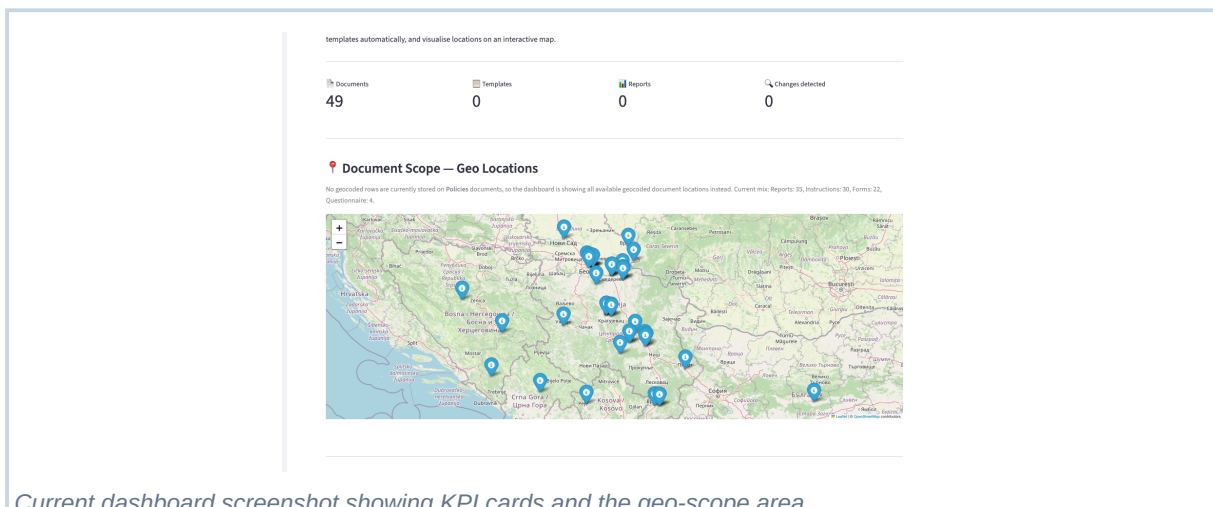
Document Scope — Geo Locations

No geocoded rows are currently stored on Policies documents, so the dashboard is showing all available geocoded document locations instead. Current mix: Reports: 35, Instructions: 30, Forms: 22, Questionnaire: 4.



Visual chapter opener generated from the current SIPMT interface

Home - Overview Dashboard



Current dashboard screenshot showing KPI cards and the geo-scope area.

Visual Focus

Current dashboard screenshot showing KPI cards and the geo-scope area.

The Home page is your workspace dashboard, providing at-a-glance metrics and overview of your data.

Key Features:

Metrics Overview

- ****Documents****: Total number of uploaded documents in the system
- ****Templates****: Number of defined report templates available
- ****Reports****: Count of generated reports from template + document combinations
- ****Changes Detected****: Number of change instances discovered via document version comparison

Document Scope Map

- Displays an interactive map of all geographic locations extracted from ****Policies**** documents
- Helps you visualize the geographic scope of your policy framework
- Requires documents to be processed with location extraction enabled
- Locations are geocoded (if `ENABLE_GEOCODING=True` in `.env`) and marked on an OpenStreetMap layer

When to Use:

- When you first open SIPMT to see workspace statistics
- To understand the geographic reach of your policies
- As a quick health check that documents are being processed

Documents - Upload & Manage Source Files

Chapter Focus

The Documents page is where you upload and manage all your source files.

Upload New Documents

Choose files (PDF, DOCX, XLSX) — max 20 MB each

Drag and drop files here
Limit 20MB per file • PDF, DOCX, DOC, XLSX, XLS

Browse files

Document category

Questionnaire

Document Library

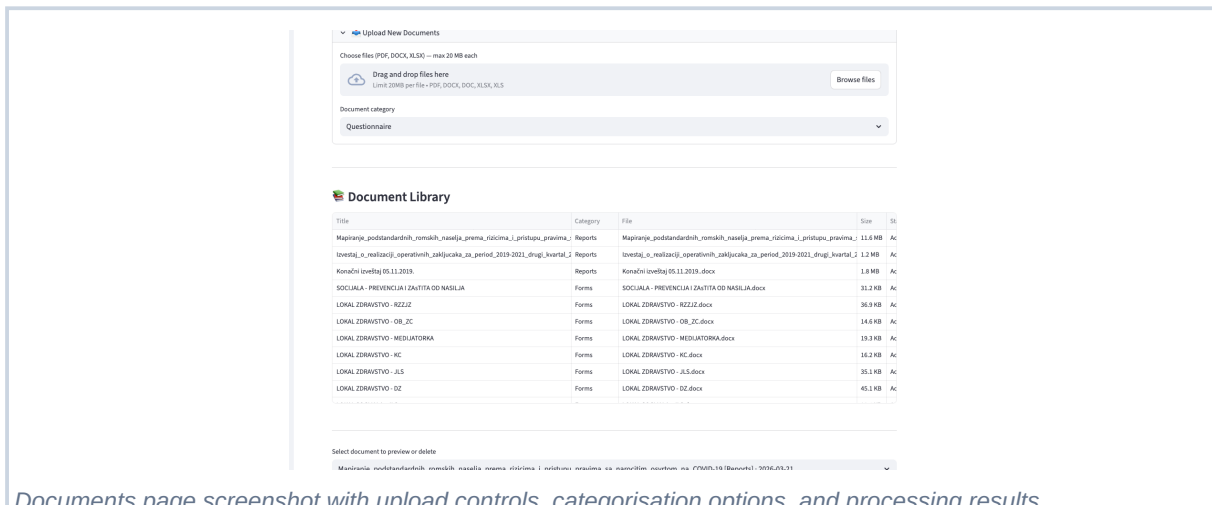
Title	Category	File	Size	Status
Mapiiranje_podstandardnih_romskih_naselja_prema_rizicima_i_pristupu_pravima_za_COVID-19 [Report] - 2020-03-31	Reports	Mapiiranje_podstandardnih_romskih_naselja_prema_rizicima_i_pristupu_pravima_za_COVID-19 [Report].pdf	11.6 MB	Active
Izvestaj_o_realizaciji_operativnih_zakljucaka_za_period_2019-2021_drugi_kvartal_2021	Reports	Izvestaj_o_realizaciji_operativnih_zakljucaka_za_period_2019-2021_drugi_kvartal_2021.pdf	1.2 MB	Active
Konačni izveštaj 05.11.2019.	Reports	Konačni izveštaj 05.11.2019.docx	1.8 MB	Active
SOCIJALA - PREVENCIJA I ZAŠTITA OD NASILJA	Forms	SOCIJALA - PREVENCIJA I ZAŠTITA OD NASILJA.docx	31.2 KB	Active
LOKAL ZDRAVSTVO - RZZJZ	Forms	LOKAL ZDRAVSTVO - RZZJZ.docx	36.9 KB	Active
LOKAL ZDRAVSTVO - OB_ZC	Forms	LOKAL ZDRAVSTVO - OB_ZC.docx	14.6 KB	Active
LOKAL ZDRAVSTVO - MEDIJATORKA	Forms	LOKAL ZDRAVSTVO - MEDIJATORKA.docx	19.3 KB	Active
LOKAL ZDRAVSTVO - KC	Forms	LOKAL ZDRAVSTVO - KC.docx	16.2 KB	Active
LOKAL ZDRAVSTVO - JLS	Forms	LOKAL ZDRAVSTVO - JLS.docx	35.1 KB	Active
LOKAL ZDRAVSTVO - DZ	Forms	LOKAL ZDRAVSTVO - DZ.docx	45.1 KB	Active

Select document to preview or delete

Maniranie podstandardnih romskih naselja prema rizicima i pristupu pravima sa narocitim osvetlom na COVID-19 [Report] - 2020-03-31

Visual chapter opener generated from the current SIPMT interface

Documents - Upload & Manage Source Files



Documents page screenshot with upload controls, categorisation options, and processing results.

Visual Focus

Documents page screenshot with upload controls, categorisation options, and processing results.

The Documents page is where you upload and manage all your source files.

Supported File Formats:

- **PDF** - Scanned documents, policy documents, reports (with or without embedded text)
- **DOCX** - Microsoft Word documents, forms, instructions
- **XLSX** - Excel spreadsheets, data tables, monitoring data

File Size & Constraints:

- **Maximum size per file**: 20 MB
- **Multiple uploads**: You can upload multiple files at once
- **Duplicate detection**: Files with identical content are automatically flagged and can be skipped

Processing Steps:

1. **Upload**: Select one or more files from your computer
2. **Categorize**: Assign each file to one of six document types:
 - **Questionnaire** - Surveys, forms for data collection
 - **Policies** - Official policy frameworks, guidelines
 - **Instructions** - How-to guides, operational manuals
 - **Forms** - Templates, application forms, compliance forms
 - **Reports** - Monitoring reports, assessment reports, evaluations
 - **Monitoring** - Ongoing tracking documents, dashboards, KPI reports
3. **Extract**: System automatically extracts:
 - **Text** - All text content from documents
 - **Tables** - Structured data formatted as CSV
 - **Named Entities** - People, organizations, locations identified via NLP
 - **Locations** - Geographic place names (countries, cities, regions)
4. **Store**: Processed content is saved to the database for analysis

Advanced Features:

- **Re-process button**: For older documents that weren't fully extracted, re-run the entire pipeline
- **View extracted text**: Click "View" next to a document to see extracted content
- **Deletion**: Remove documents you no longer need (soft delete)

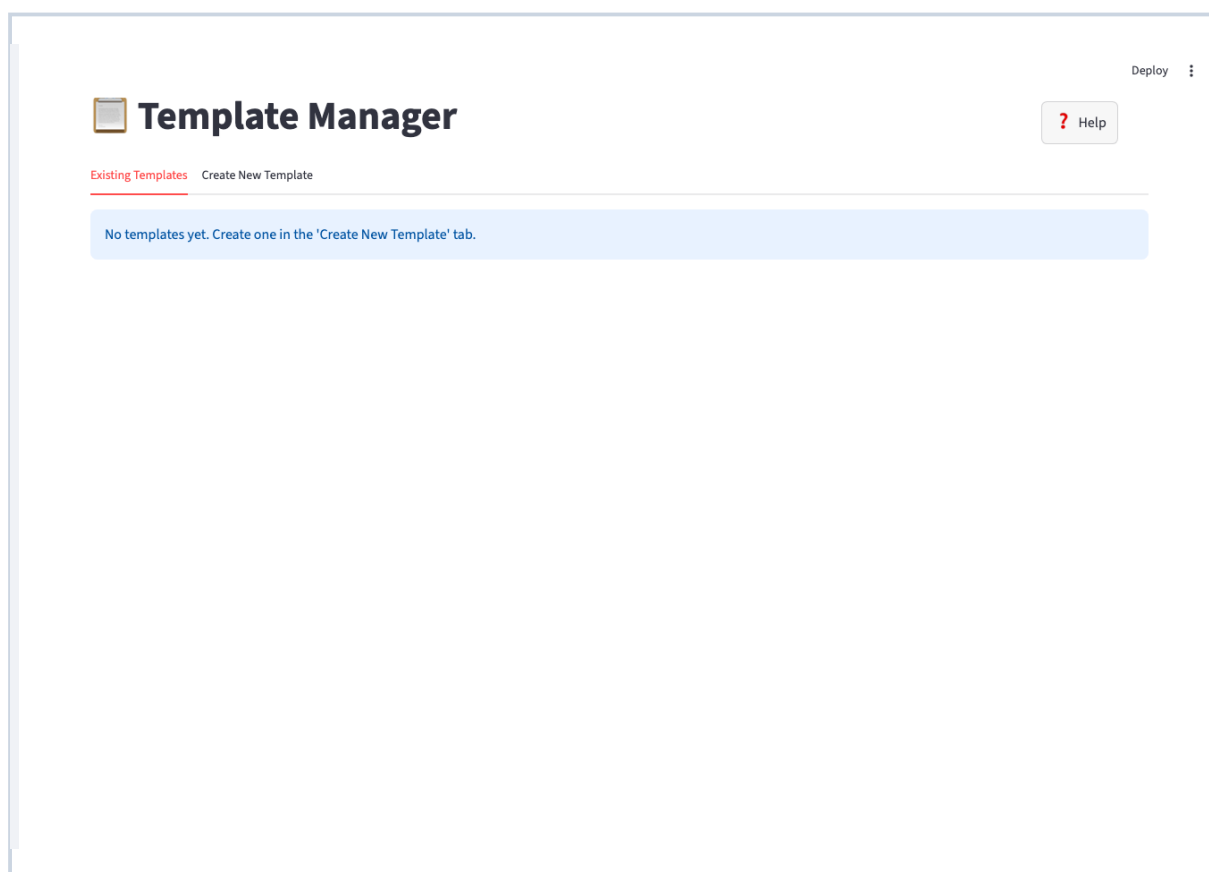
When to Use:

- To onboard new documents for analysis
- To prepare documents for report generation or compliance checking
- To update your data scope with new policy versions

Templates - Define Data Extraction Fields

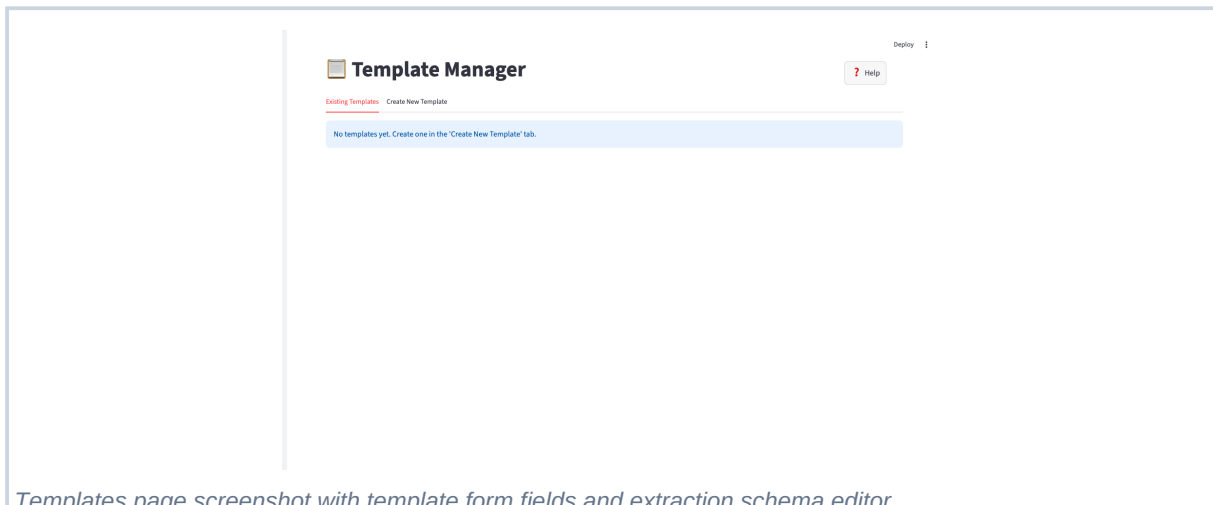
Chapter Focus

The Templates page lets you create reusable field definitions for automated data extraction. A report template is a structured specification of the information you want to extract from documents: Once you create a template, SIPMT uses fuzzy matching to automatically find corresponding values in documents:



Visual chapter opener generated from the current SIPMT interface

Templates - Define Data Extraction Fields



Templates page screenshot with template form fields and extraction schema editor.

Visual Focus

Templates page screenshot with template form fields and extraction schema editor.

The Templates page lets you create reusable field definitions for automated data extraction.

What is a Template?

A report template is a structured specification of the information you want to extract from documents:

- A **name** that describes what the template captures (e.g., "Social Inclusion Indicators", "Health Policy Compliance Check")
- A **set of fields**, each with:
 - **Field name** - A descriptive label (e.g., "Target Population", "Implementation Date")
 - **Field type** - The data type SIPMT will look for:
 - **Text** - Free-form text fields (e.g., description, summary)
 - **Number** - Numeric values (e.g., budget amount, percentage)
 - **Date** - Calendar dates (e.g., effective date, review date)
 - **Boolean** - Yes/No fields (e.g., "Is monitoring required?")
 - **List** - Comma-separated or bulleted values (e.g., stakeholder names)
 - **Location** - Place names, regions, countries
 - **Description** (optional) - Context to help the extractor
 - **Priority** - Whether field is Required or Optional

Template Matching Process:

Once you create a template, SIPMT uses fuzzy matching to automatically find corresponding values in documents:

1. Field name is compared against extracted document text
2. Semantic analyzer (Gemini or Ollama) helps interpret intent
3. Matched values are auto-filled in generated reports

Creating a Template:

1. Click **"Create New Template"**

2. Enter template name
3. Add fields using the table widget:
 - Specify field name, type, description
 - Mark Required/Optional
4. Click "Save Template"

Use Cases:

- Define fields for different document types (policies, reports, questionnaires)
- Create institution-specific templates for consistent data extraction
- Enable standardized reporting across multiple documents

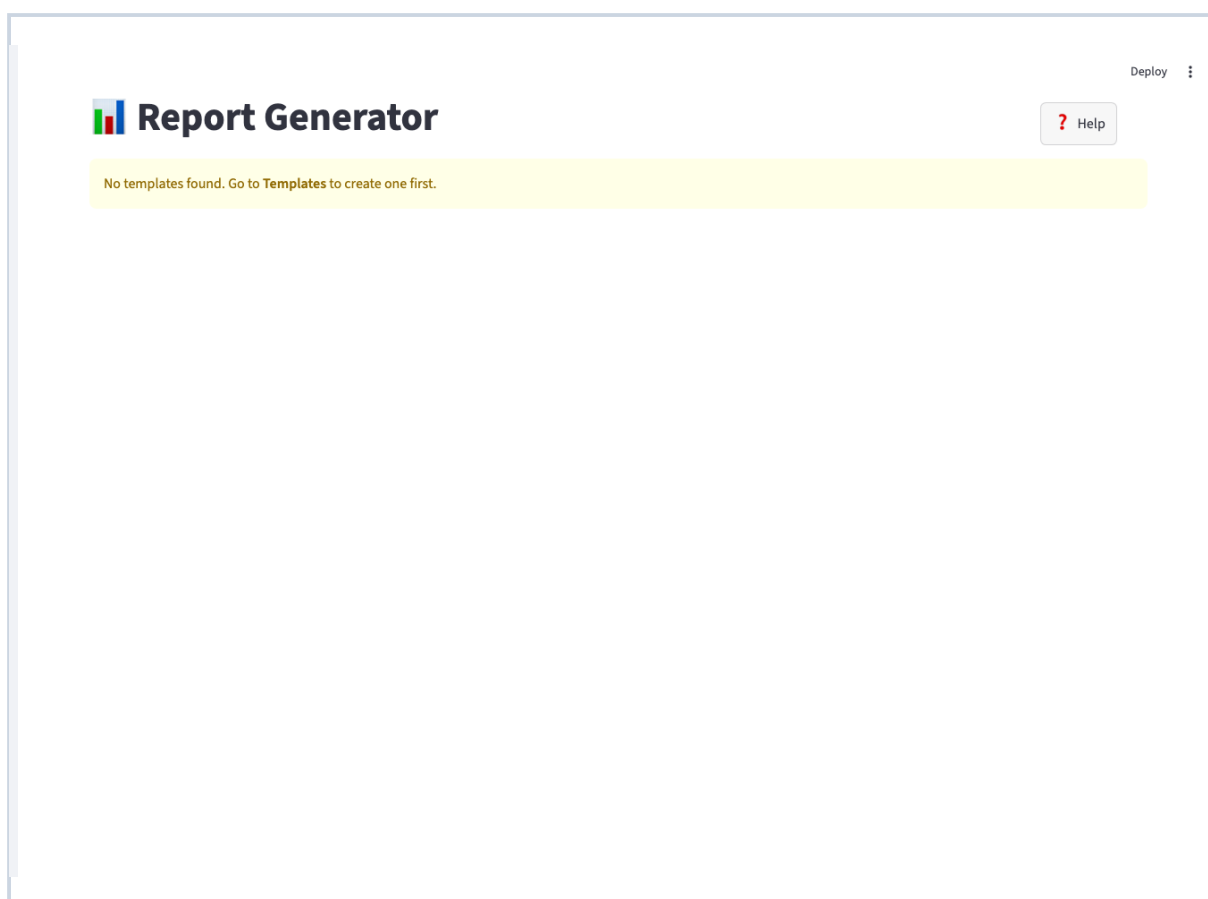
When to Use:

- Before generating reports - define the structure first
- When you want consistent data extraction across similar documents
- To standardize information collection

Reports - Generate Structured Data Extracts

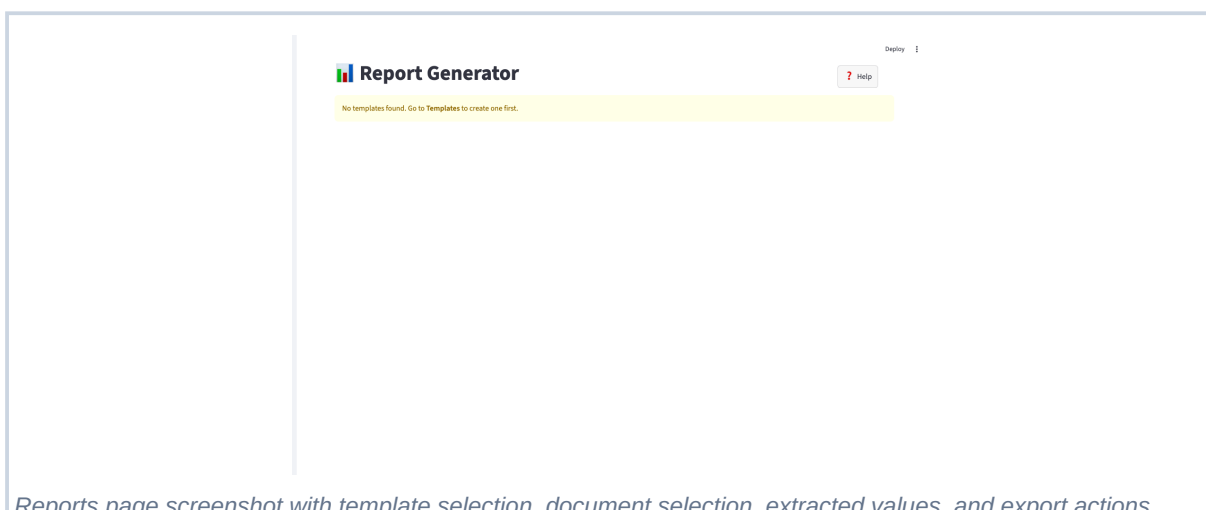
Chapter Focus

The Reports page handles automated extraction of structured data using templates.



Visual chapter opener generated from the current SIPMT interface

Reports - Generate Structured Data Extracts



Visual Focus

Reports page screenshot with template selection, document selection, extracted values, and export actions.

The Reports page handles automated extraction of structured data using templates.

The Report Generation Workflow:

1. **Select Template** - Choose which template defines the extraction fields
2. **Select Documents** - Pick one or more source documents to extract from
3. **Run Extraction** - System processes the documents with fuzzy matching + semantic analysis
4. **Review Results** - View extracted values, confidence scores, and any missing fields
5. **Export Report** - Save as CSV, JSON, or Excel for downstream analysis

Extraction Confidence:

- Each extracted field includes a **confidence score** (0-100%)
- **High confidence** (>80%) - Field was clearly identified in the source
- **Medium confidence** (50-80%) - Field found but with some uncertainty
- **Low confidence** (<50%) - Weak match; manual review recommended
- **Missing** - Field not found in any of the selected documents

Feature Highlights:

- **Batch extraction** - Process multiple documents at once against same template
- **Field review** - Manually edit extracted values before export
- **Audit trail** - Each report captures which documents were used and when
- **Storage** - All generated reports are saved to the database
- **Export formats** - CSV (spreadsheet), JSON (system integration), PDF (sharing)

When to Use:

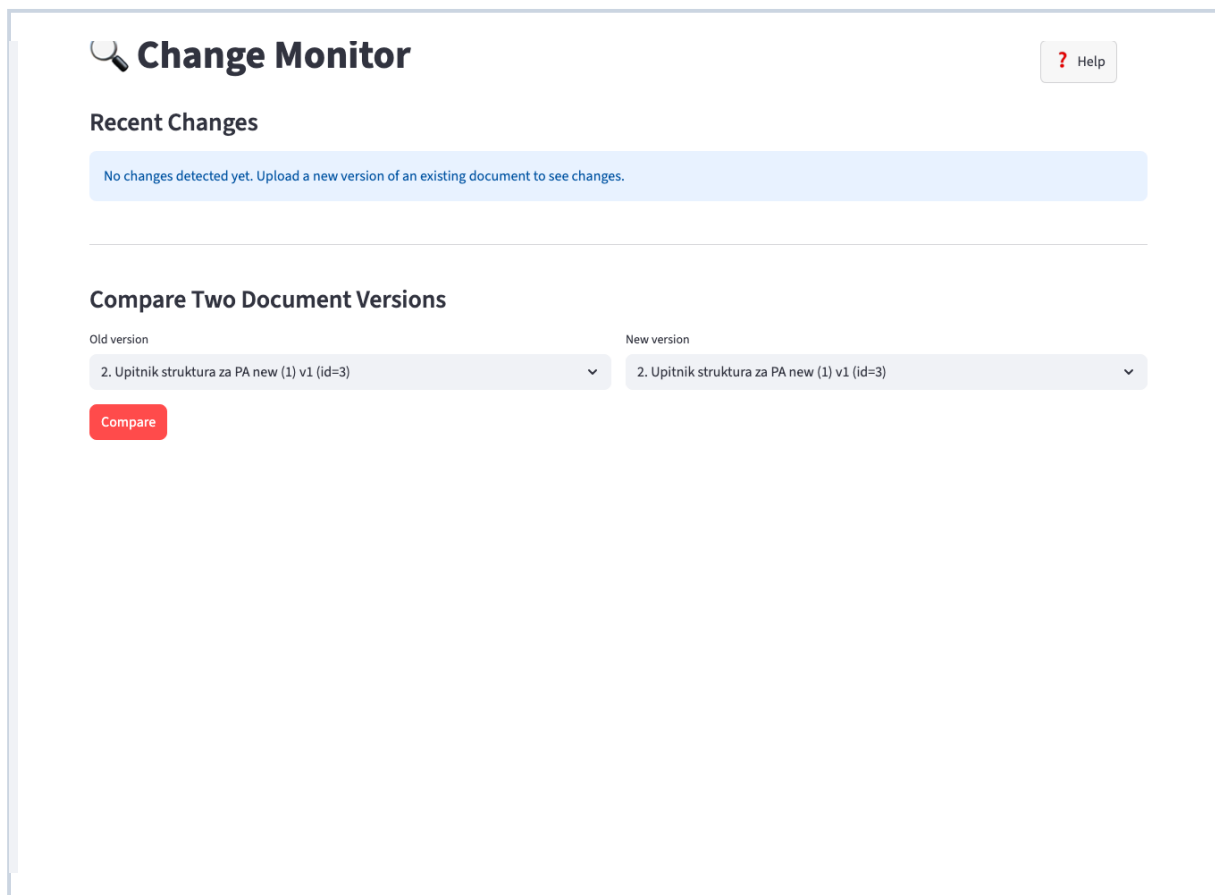
- To fill standardized reporting forms automatically
- To compile consistent data from many similar documents
- To create CSVs for further analysis or visualization

- When you need verified extracted values with confidence scores

Changes - Track Document Versions

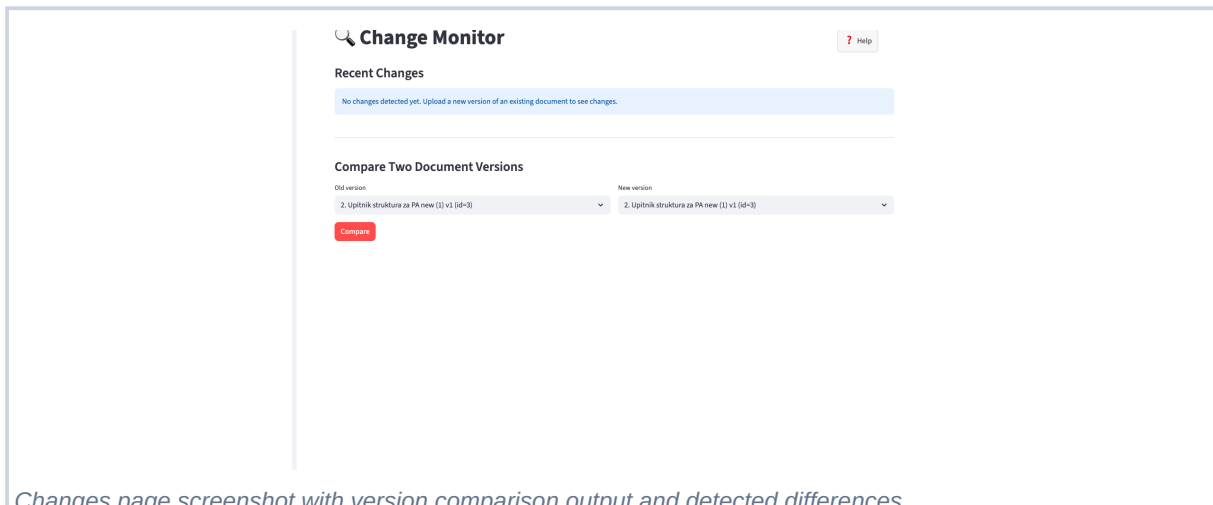
Chapter Focus

The Changes page detects and visualizes differences between document versions. SIPMT detects it has the same name but different content



Visual chapter opener generated from the current SIPMT interface

Changes - Track Document Versions



Changes page screenshot with version comparison output and detected differences.

Visual Focus

Changes page screenshot with version comparison output and detected differences.

The Changes page detects and visualizes differences between document versions.

Change Tracking Workflow:

1. **Automatic Detection** - When you upload a revised version of an existing document, SIPMT detects it has the same name but different content
2. **Diff Generation** - The system runs a comparison to identify:
 - **Added** - New sections, paragraphs, or content
 - **Removed** - Sections deleted in the new version
 - **Modified** - Text that changed but wasn't added/removed
3. **Visualization** - Side-by-side display with color highlighting:
 - **Green** - Added content
 - **Red** - Removed content
 - **Yellow** - Modified content

Change Audit:

- **Timestamp** - Exactly when each version was uploaded
- **Size delta** - How many bytes changed between versions
- **Persistent storage** - All detected changes are saved for compliance auditing

Use Cases:

- Track policy evolution over time
- Monitor when instructions or procedures have been updated
- Ensure you're aware of all changes to reference documents
- Create a versioning audit trail for compliance reporting

When to Use:

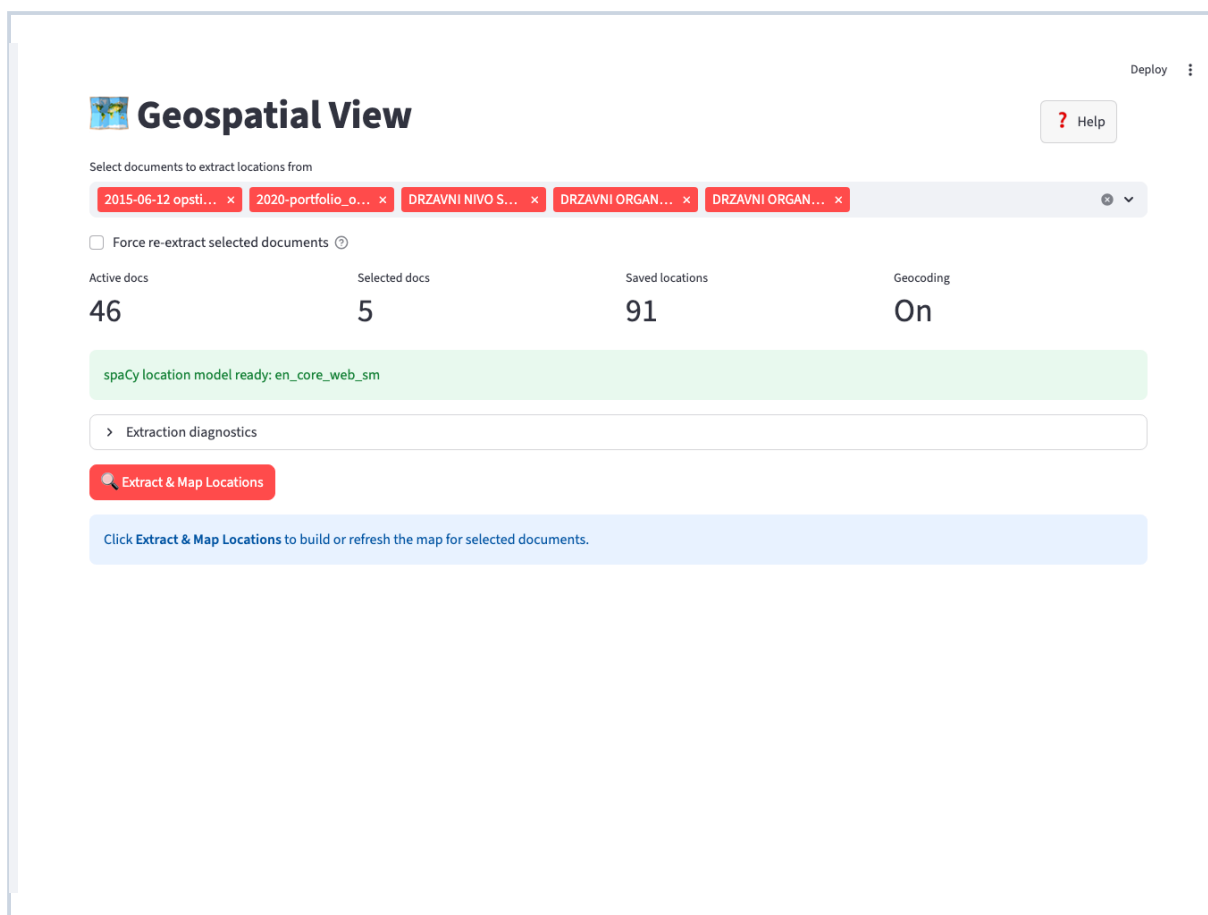
- When you upload a new version of an existing document
- To review what changed in a policy update

- For compliance audits that require change tracking
- Before adopting a new document version

Map - Geospatial Visualization

Chapter Focus

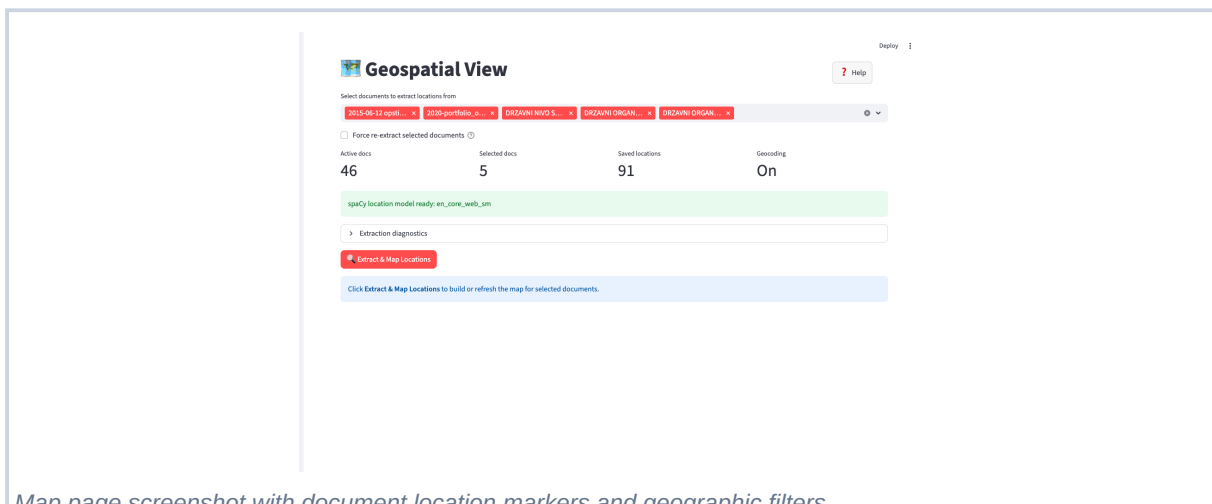
The Map page provides interactive geographic visualization of extracted locations.



The screenshot displays the 'Geospatial View' interface. At the top right, there is a 'Deploy' button and a 'Help' button. Below the title, a section 'Select documents to extract locations from' shows five document thumbnails: '2015-06-12 opsti...', '2020-portfolio_o...', 'DRZAVNI NIVO S...', 'DRZAVNI ORGAN...', and 'DRZAVNI ORGAN...'. A checkbox 'Force re-extract selected documents' is present. A summary table shows: Active docs (46), Selected docs (5), Saved locations (91), and Geocoding (On). A green status bar indicates 'spaCy location model ready: en_core_web_sm'. Below this is an 'Extraction diagnostics' section. A red button 'Extract & Map Locations' is prominently displayed. At the bottom, a blue instruction box states: 'Click Extract & Map Locations to build or refresh the map for selected documents.'

Visual chapter opener generated from the current SIPMT interface

Map - Geospatial Visualization



Map page screenshot with document location markers and geographic filters.

Visual Focus

Map page screenshot with document location markers and geographic filters.

The Map page provides interactive geographic visualization of extracted locations.

How It Works:

- Location Extraction** - When documents are processed, place names are identified using:
 - Pattern matching** - Recognizes known country/region strings
 - NLP (spaCy)** - Named Entity Recognition for place mentions
- Geocoding** (optional):
 - If `ENABLE_GEOCODING=True` in `.env`, each place name is converted to latitude/longitude
 - Uses **Nominatim** (OpenStreetMap's free geocoding service)
 - Requires internet connectivity
- Visualization** - Interactive map shows:
 - Markers** - Each location pinned on OpenStreetMap
 - Pop-ups** - Click marker to see document source and context
 - Zoom/Pan** - Explore the geographic scope interactively

Map Features:

- Filter by document type** - Show only locations from Policies, Reports, Monitoring, etc.
- Zoom to region** - Focus on a specific geographic area
- Export map** - Save as HTML or screenshot for presentations

Configuration:

- Enable geocoding**: Set `ENABLE_GEOCODING=True` in your `.env` file
- Select documents**: Choose which documents to visualize
- Update locations**: Re-process documents to extract new locations

Use Cases:

- Visualize coverage area of policies and programs
- Identify geographic gaps in implementations
- Share scope with stakeholders in an interactive format
- Plan regional expansion or resource allocation

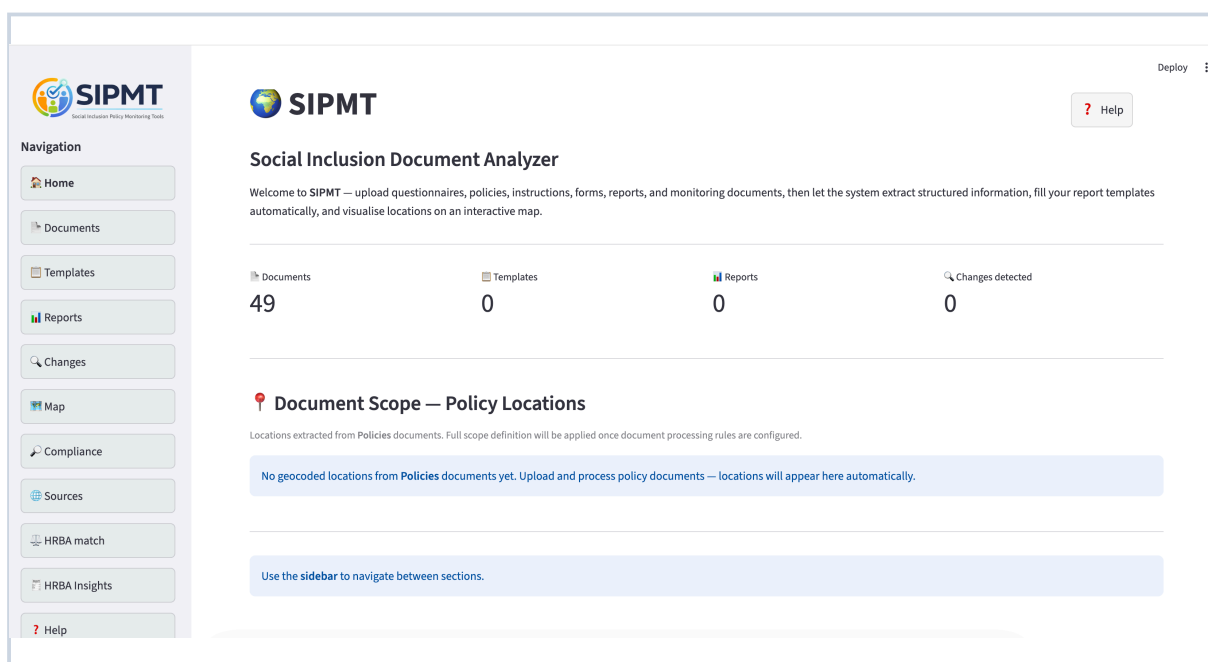
When to Use:

- After uploading Policies or Programs documents
- To brief stakeholders on geographic scope
- For regional analysis and planning
- In presentations or reports to show coverage

KPIs & Charts - Definitions, Data Sources, and Population Details

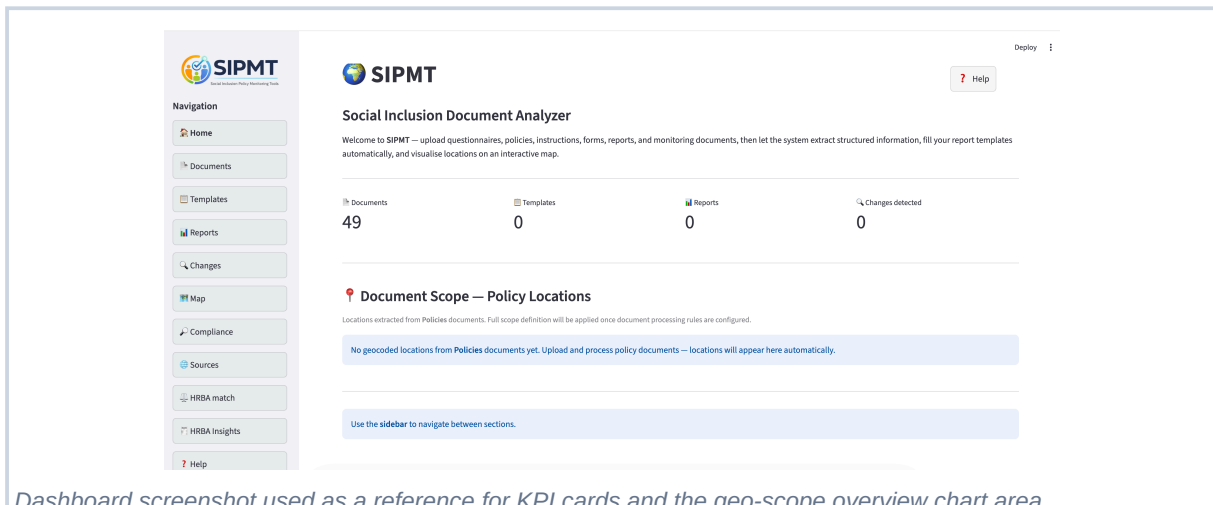
Chapter Focus

This section explains every KPI shown on the Home dashboard and every chart used across the application: what each metric represents, which DB tables and queries populate it, how often it updates, and any caveats. General notes Dashboard KPIs (Home page)



Visual chapter opener generated from the current SIPMT interface

KPIs & Charts - Definitions, Data Sources, and Population De



Dashboard screenshot used as a reference for KPI cards and the geo-scope overview chart area.

Visual Focus

Dashboard screenshot used as a reference for KPI cards and the geo-scope overview chart area.

This section explains every KPI shown on the Home dashboard and every chart used across the application: what each metric represents, which DB tables and queries populate it, how often it updates, and any caveats.

General notes

- All KPIs on the dashboard are read-only aggregates run against the documents, report_templates, reports, and document_changes tables.
- Charts may be rendered from aggregated SQL queries or from in-memory analysis payloads saved in JSON columns (for semantic analyses and reports).
- Dashboard values are computed at render time; they reflect the current contents of the database when the page is loaded or when the user navigates to the page.

Dashboard KPIs (Home page)

- **Documents** (label: " Documents")
 - What it represents: Total count of document records stored for the current organisation.
 - Data source / SQL: `SELECT COUNT(*) FROM documents WHERE organisation_id = ?`
 - How it's created: incremented when a new document is uploaded and processed; deletions reduce the count (soft-deletes respect status field).
 - Update frequency: real-time at page render.
- **Templates** (label: " Templates")
 - What it represents: Number of saved report templates available for extraction & reporting.
 - Data source / SQL: `SELECT COUNT(*) FROM report_templates`
 - How it's created: created via the Templates page when the user saves a new template.
 - Caveats: template bodies may include large JSON; the KPI counts templates regardless of body size or source (DB or file-imported).
- **Reports** (label: " Reports")
 - What it represents: Count of generated reports (structured extraction outputs) stored in

reports table.

- Data source / SQL: `SELECT COUNT(*) FROM reports WHERE organisation_id = ?`
- How it's created: when a user runs extraction against documents and saves or exports results.
- ****Changes detected**** (label: "Changes detected")
- What it represents: Number of detected change instances (diff records) between document versions.
- Data source / SQL: `SELECT COUNT(*) FROM document_changes WHERE organisation_id = ?`
- How it's created: when a new upload is identified as a new version of an existing document and the diff process creates change rows.

Map & Location Data

- The scope map shows geocoded locations joined to documents (see locations table).
- Primary SQL used for the dashboard map (Policies-preferred):
- Policies restricted: `SELECT` rows from locations JOIN documents WHERE document_type = 'Policies' AND geocoded = 1.
- Fallback: all geocoded locations when Policies have none.
- How markers are populated: each geocoded location row provides latitude/longitude, place_name, context and document_title used in the pop-up.

Compliance page charts & KPIs

- ****Compliance Score****
- What it represents: a normalized alignment score (0-1) computed by ComplianceChecker combining keyword, requirement, and section coverage with small body/category bonuses.
- How it's computed: see `_score_reference_template` and `_build_executive_summary` in the compliance page code - the algorithm combines token-coverage metrics and applies weights (keywords 45%, requirements 35%, sections 20% plus bonuses).
- Source: analysis payloads stored in `semantic_analyses` and the temporary result in-memory while running the analysis.
- ****SHAP-style heatmap (feature contribution proxy)****
- What it represents: an explainability proxy that shows which factors (Present elements, Strengths, Missing, Partial, Gaps, Recommendation pressure, Keyword/Requirement/Section coverage) contributed positively or negatively to the final compliance score.
- How it's created: `_compute_shap_proxy` aggregates counts from the analysis result and returns normalized contribution values between -1 and +1.
- Chart population: uses Plotly Heatmap with a single-row z array of contributions and an explicit color scale centered at 0.
- ****Classification & Comparison score histories (line/points)****
- What they represent: time-series of the classifier confidence and template comparison scores for the current monitor session or historical analyses.
- Source: `classification_confidence_history` and `comparison_score_history` arrays maintained in the compliance monitor state; persisted analyses are available via `checker.get_analyses(document_id)`.

-
- How populated: appended as the monitor runs each stage (classify, match, compare, persist). Charts use Plotly scatter or line traces with points recorded in *_points lists.
 - **Gantt-style timeline (per-segment timeline)**
 - What it represents: per-segment generation timing when streaming LLM outputs; bars are sized by score and colored by AAAQ label.
 - Source: streaming chunks processed during Ollama/Gemini runs, each tracked with start/end and assigned category/score.
 - How populated: compliance monitor records chunk timestamps and scores into chunk_history; the timeline is rendered from those records.
-

HRBA page visuals

- **Live streaming preview**: incremental text from the LLM while running a HRBA analysis. Populated by streaming HTTP responses from Ollama (if supported) or replaced by final JSON when streaming not available.
 - **Per-segment Gantt**: same mechanism as Compliance timeline - segments are generated and timed during model runs.
-

Other charts across the app

- **Template source health table**: a diagnostics table produced by _probe_source_url_health that verifies external source URLs for templates; populated by issuing HEAD/GET requests and collecting HTTP status and suggestions.
 - **Map exports & filters**: map layers are created by MapGenerator.build_map() using the scope_locations rows; filters are executed by SQL before the map is built.
-

Troubleshooting & caveats

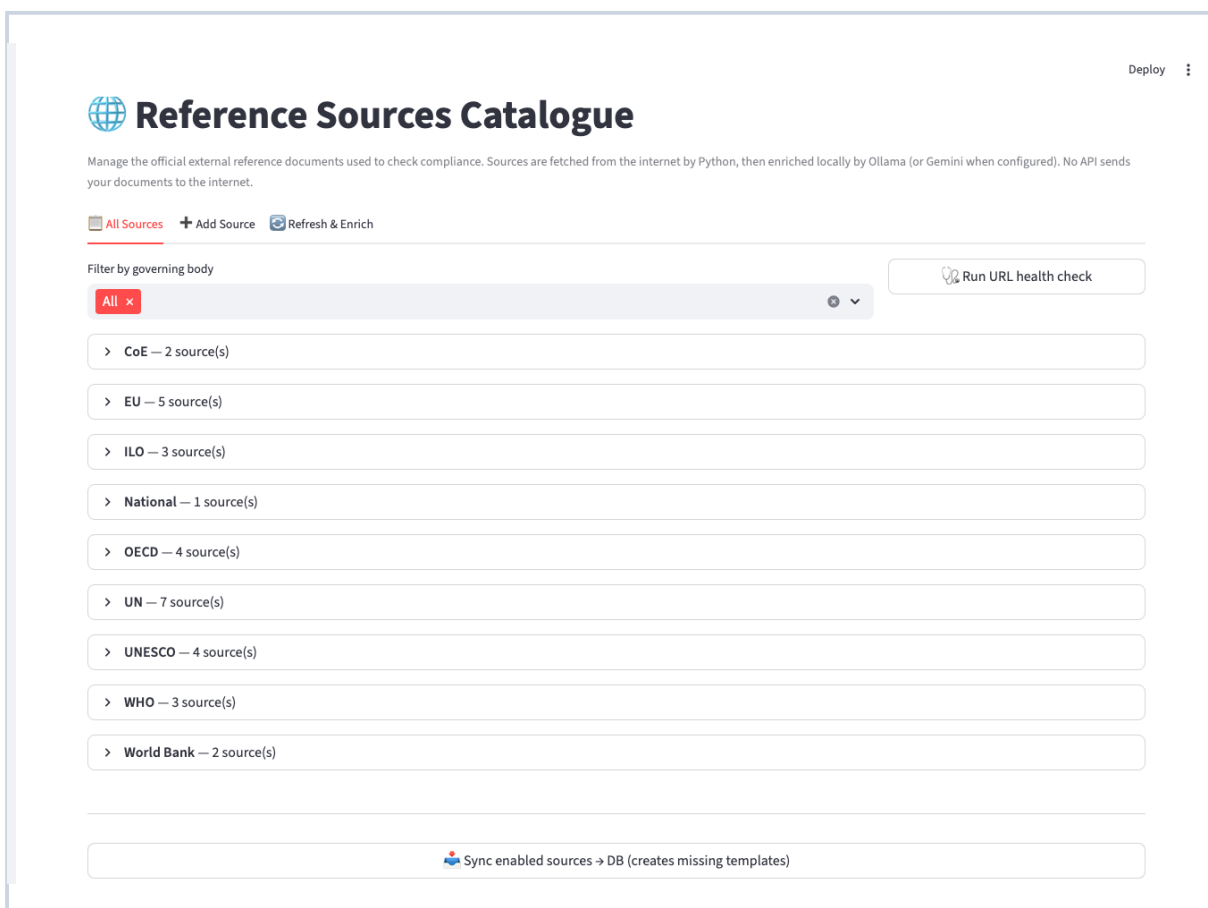
- If a KPI is unexpectedly zero or stale, check whether your organisation filter (organisation_id) is set and whether documents have status='active'.
- Semantic analysis-derived charts rely on cached analysis payloads; re-run analyses or clear cache if you believe stale results are shown.
- Charts that rely on LLM streaming require the selected provider to support streaming; otherwise charts will update once the final response arrives.

CHAPTER 08

Sources - Reference Catalogue & Refresh Workflow

Chapter Focus

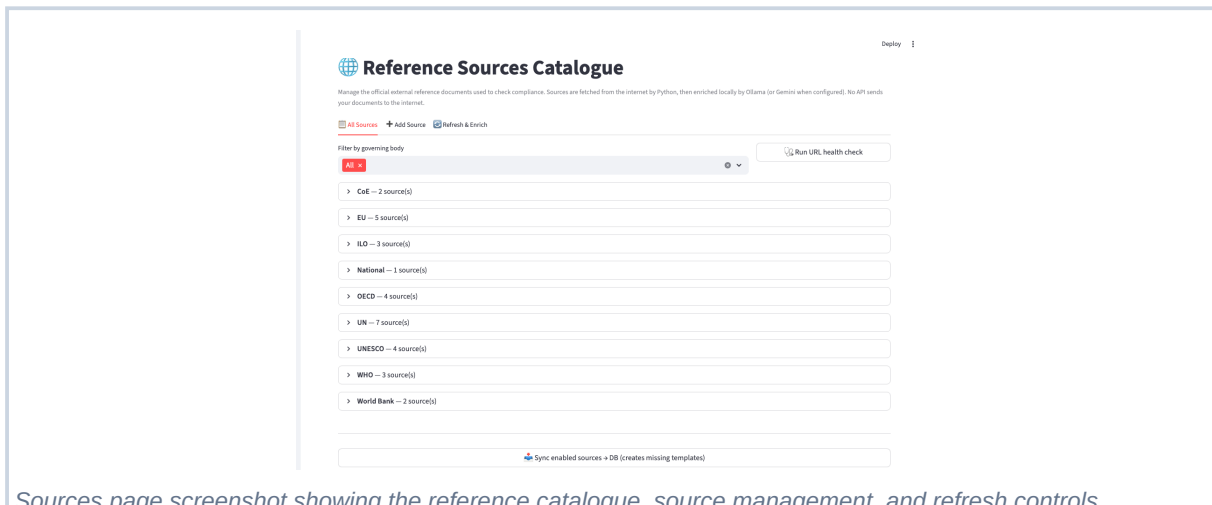
The Sources page manages the official external reference material used by the compliance engine.



The screenshot displays the 'Reference Sources Catalogue' interface. At the top right, there is a 'Deploy' button with a dropdown arrow. The main heading is 'Reference Sources Catalogue' with a globe icon. Below the heading, a descriptive text states: 'Manage the official external reference documents used to check compliance. Sources are fetched from the internet by Python, then enriched locally by Ollama (or Gemini when configured). No API sends your documents to the internet.' Below this text are two buttons: 'All Sources' (highlighted in red) and '+ Add Source'. To the right of these buttons is a 'Refresh & Enrich' button with a circular arrow icon. A 'Filter by governing body' section follows, featuring a search bar with 'All' selected and a dropdown arrow. To the right of the filter is a 'Run URL health check' button with a checkmark icon. Below the filter, a list of governing bodies is shown, each with a right-pointing arrow and the number of sources: CoE — 2 source(s), EU — 5 source(s), ILO — 3 source(s), National — 1 source(s), OECD — 4 source(s), UN — 7 source(s), UNESCO — 4 source(s), WHO — 3 source(s), and World Bank — 2 source(s). At the bottom, there is a button labeled 'Sync enabled sources → DB (creates missing templates)' with a sync icon.

Visual chapter opener generated from the current SIPMT interface

Sources - Reference Catalogue & Refresh Workflow



Sources page screenshot showing the reference catalogue, source management, and refresh controls.

Visual Focus

Sources page screenshot showing the reference catalogue, source management, and refresh controls.

The Sources page manages the official external reference material used by the compliance engine.

What This Page Does:

- Lists all configured source catalog entries and reference bodies
- Lets you add new official URLs for standards, directives, and guidance notes
- Runs refresh workflows to fetch source pages and update the internal template database
- Shows a current database snapshot of the active reference set

Main Interface Elements:

- **Catalogue table** - Displays the current source records, grouped by body and category
- **Add new source form** - Creates a new source entry with body, label, URL, and file hint
- **Fetch / enrich controls** - Pulls source content from the internet, then enriches it with the selected provider
- **DB snapshot section** - Shows the currently materialized template records used by compliance analysis

When to Use:

- When new international or institutional guidance needs to be added
- When existing external URLs changed and references must be refreshed
- Before running compliance analyses that depend on newly updated source material

Compliance - Semantic Analysis Against Standards

Chapter Focus

The Compliance page provides advanced AI-powered semantic analysis of your documents against international reference frameworks.



Semantic Compliance Analysis

Deploy

? Help

Refresh health

OLLAMA not probed yet — click [Refresh health](#) to check availability.



Ollama Tuning Diagnostics

Active runtime options and recent timing snapshot for local tuning.

Runtime Options

Option	Value
Provider	OLLAMA
Model	unknown
OLLAMA_TIMEOUT_SEC	600
OLLAMA_KEEP_ALIVE	60m
OLLAMA_NUM_CTX	4096
OLLAMA_NUM_THREAD	auto
OLLAMA_NUM_GPU	auto
OLLAMA_NUM_BATCH	512

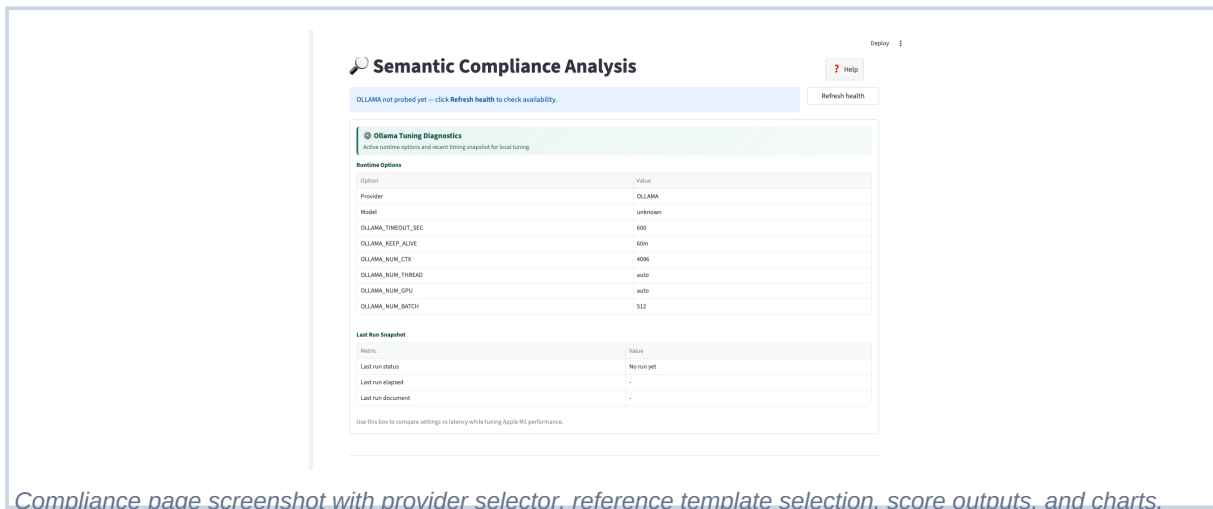
Last Run Snapshot

Metric	Value
Last run status	No run yet
Last run elapsed	-
Last run document	-

Use this box to compare settings vs latency while tuning Apple M1 performance.

Visual chapter opener generated from the current SIPMT interface

Compliance - Semantic Analysis Against Standards



Compliance page screenshot with provider selector, reference template selection, score outputs, and charts.

Visual Focus

Compliance page screenshot with provider selector, reference template selection, score outputs, and charts.

The Compliance page provides advanced AI-powered semantic analysis of your documents against international reference frameworks.

Semantic Analysis Workflow:

- Provider Selection** - Choose your AI backend:
 - Google Gemini** (cloud, requires API key) - Advanced reasoning, internet connectivity
 - Ollama** (local, self-hosted) - Privacy-preserving, offline operation
- Health Check** - Verify the selected AI provider is reachable and working
- Document Selection** - Choose which document to analyze
- Reference Template** - Select an international framework (UN, UNESCO, EU, etc.)
- Run Analysis** - AI compares document against reference framework
- Review Report** - Structured compliance assessment with recommendations

Reference Frameworks Available:

- UN Human Rights** - Universal Declaration of Human Rights frameworks
- UN Sustainable Development Goals (SDGs)** - Goal 10 (Reduced Inequalities) focused
- UNESCO Education** - Inclusive education standards and guidelines
- EU Equality Directive** - EU equality and non-discrimination standards
- EU Social Inclusion** - Social cohesion and inclusion directives

Compliance Report Structure:

- Document Classification** - AI categorizes the document type/scope
- Compliance Score** - 0-100% alignment with reference framework
- Present Elements** - Which requirements are already addressed
- Identified Gaps** - Missing or underspecified areas
- Recommendations** - Actionable steps to improve compliance
- Source References** - Links to relevant external standards

Advanced Features:

- **Template Version History** - Admin view of reference template updates and lineage
- **Refresh Sources** - Manually update reference templates from external sources
- **Error Transparency** - Clear error messages if provider is down

Model Configuration:

- **Gemini**: Requires GEMINI_API_KEY in .env, set SEMANTIC_LLM_PROVIDER=gemini
- **Ollama**: Requires Ollama service running locally, set SEMANTIC_LLM_PROVIDER=ollama
- **Model selection**: Configure via OLLAMA_MODEL in .env (default: qwen2.5:14b-instruct)

When to Use:

- To assess policy alignment with international standards
- For compliance reporting and auditing
- To identify gaps in your social inclusion framework
- Before finalizing new policies or procedures
- For stakeholder reporting on standards compliance

Configuration - Environment Setup

Chapter Focus

Document Processing Semantic Analysis Geospatial

Visual chapter opener generated from the current SIPMT interface

Configuration - Environment Setup

Key Configuration Options (in .env file)

Document Processing

- UPLOAD_FOLDER - Directory where uploaded files are stored
- ALLOWED_EXTENSIONS - File types to accept (pdf, docx, xlsx)
- MAX_FILE_SIZE_MB - Server-side file size limit
- ENABLE_OCR - Extract text from scanned/image documents

Semantic Analysis

- ENABLE_SEMANTIC_ANALYSIS - Turn semantic compliance analysis on/off
- SEMANTIC_LLM_PROVIDER - Choose gemini or ollama
- GEMINI_API_KEY - Your Google Gemini API key (for cloud analysis)
- GEMINI_MODEL - Model name (default: gemini-2.0-flash)
- OLLAMA_BASE_URL - Ollama service endpoint (default: http://localhost:11434)
- OLLAMA_MODEL - Local model to use (default: qwen2.5:14b-instruct)

Geospatial

- ENABLE_GEOCODING - Enable automatic geocoding of locations (requires internet)

Logging

- LOG_FILE - Path to application logs
- LOG_LEVEL - Verbosity (DEBUG, INFO, WARNING, ERROR)

Getting Started with Different Providers:

Using Google Gemini

```
SEMANTIC_LLM_PROVIDER=gemini  
GEMINI_API_KEY=your-api-key-here  
GEMINI_MODEL=gemini-2.0-flash
```

[Get API key](<https://ai.google.dev>)

Using Ollama (Local)

```
SEMANTIC_LLM_PROVIDER=ollama  
OLLAMA_BASE_URL=http://localhost:11434  
OLLAMA_MODEL=qwen2.5:14b-instruct
```

[Ollama Installation](<https://ollama.ai>)

Then pull the model:

```
bash
```

```
ollama pull qwen2.5:14b-instruct
```

CHAPTER 11

HRBA - AAAQ Matching & Insights

Chapter Focus

The HRBA page scans documents for AAAQ indicators (Availability, Accessibility, Acceptability, Quality) using either a fast spaCy matcher or the local Ollama LLM. Key features:



HRBA — AAAQ Matcher

Deploy

? Help

Use the HRBA matcher to scan text for AAAQ indicators (Availability, Accessibility, Acceptability, Quality). Select documents from the database for batch analysis or paste text below.

Matcher

☒ spaCy (fast)
 ☐ Ollama LLM (JSON)

Select documents to analyze

Choose options

Analyze selected

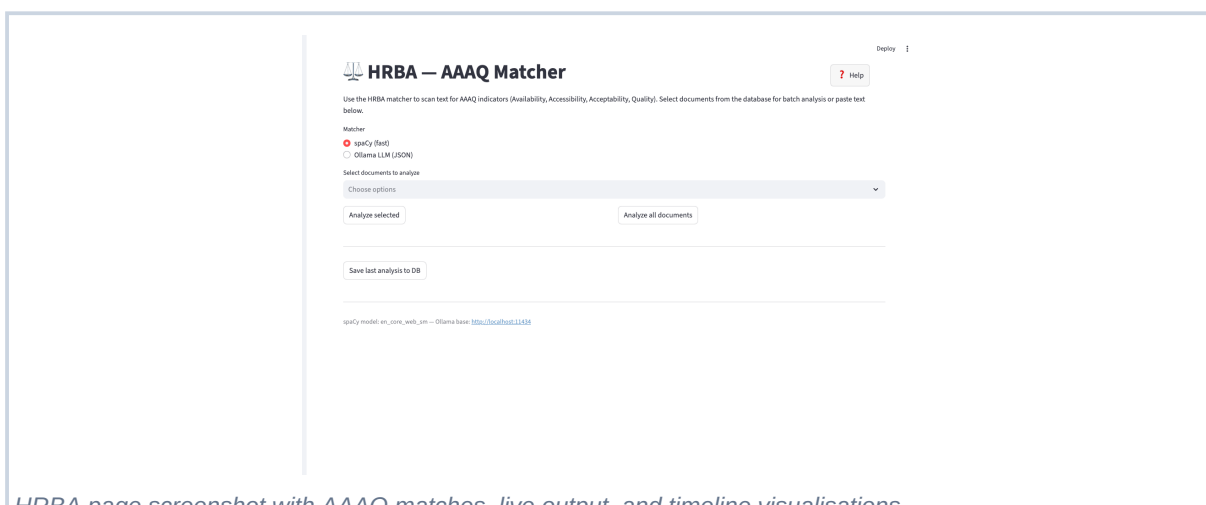
Analyze all documents

Save last analysis to DB

spaCy model: en_core_web_sm — Ollama base: <http://localhost:11434>

Visual chapter opener generated from the current SIPMT interface

HRBA - AAAQ Matching & Insights



HRBA page screenshot with AAAQ matches, live output, and timeline visualisations.

Visual Focus

HRBA page screenshot with AAAQ matches, live output, and timeline visualisations.

The HRBA page scans documents for AAAQ indicators (Availability, Accessibility, Acceptability, Quality) using either a fast spaCy matcher or the local Ollama LLM.

Key features:

- **Live streaming generation**: When using Ollama the app displays a live text preview of the model's output in a right-side pane while the model is generating.
- **Per-segment timeline (Gantt)**: The page records per-segment start/end times during streaming and renders a Gantt-style timeline showing when each segment was generated. Bars are sized by score and colored by the highest-scoring AAAQ category.
- **Save analyses**: Store full JSON justifications into semantic_analyses for later review on the HRBA Insights page.

How to interpret the UI:

- The **live preview** shows incremental text as the model streams. If the model returns only a final JSON object, the preview will update when the final content arrives.
- The **timeline** visualises per-document generation events. Longer bars indicate higher scoring results (length is proportional to score). Use the timeline to spot slow segments or clustering of analyses by document.

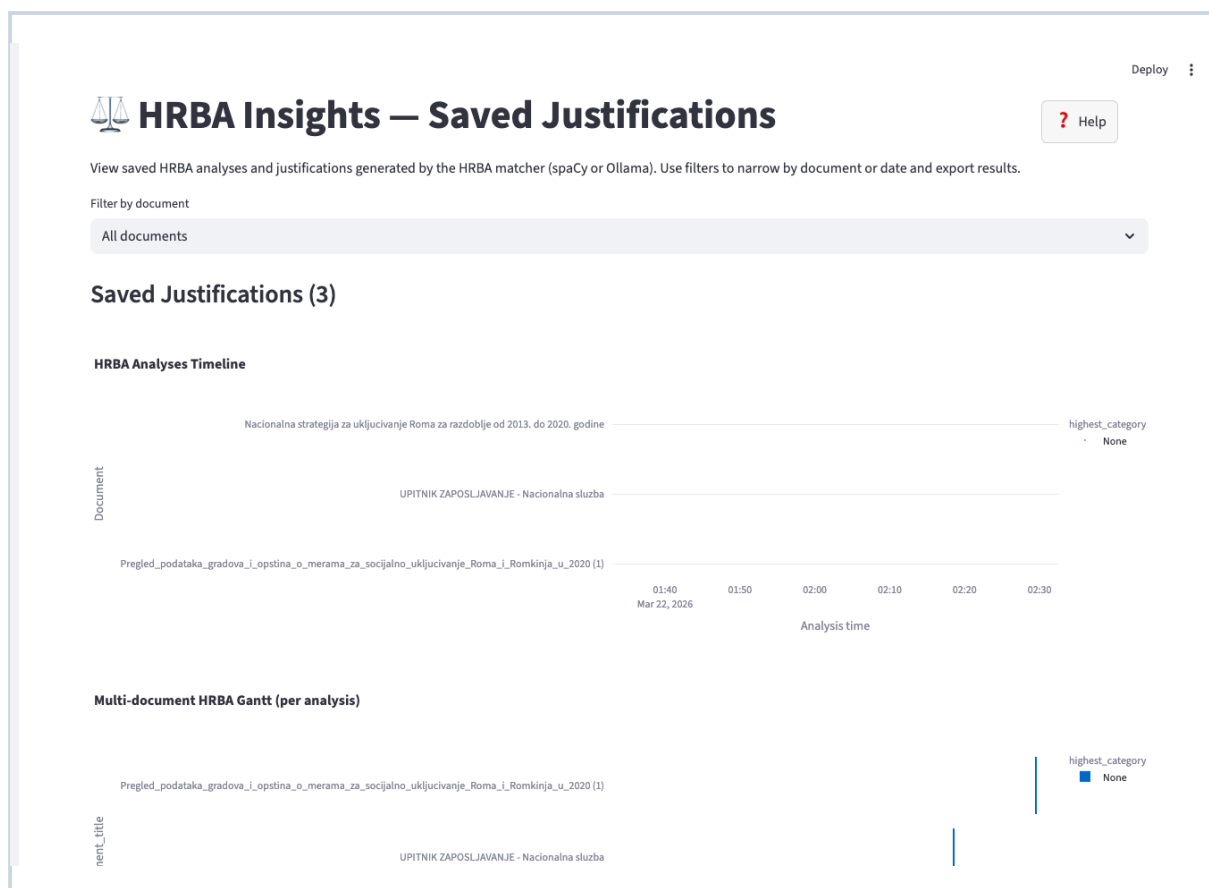
Troubleshooting:

- If you see no streaming chunks, ensure OLLAMA_BASE_URL and OLLAMA_MODEL are correct and the Ollama service is running. Some models may not stream intermediate fragments.
- For long-running generations, increase OLLAMA_TIMEOUT in your environment (seconds).
- If generation appears slow, warm the model or restart the Ollama process.

HRBA Insights - Saved Justifications & Review

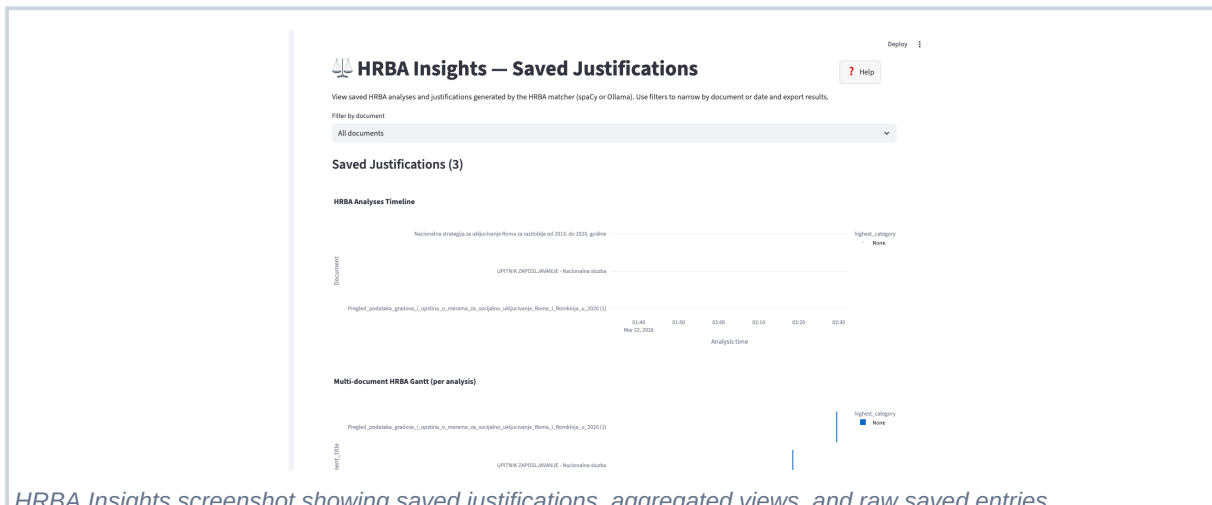
Chapter Focus

The HRBA Insights page is the review workspace for previously saved AAAQ analyses.



Visual chapter opener generated from the current SIPMT interface

HRBA Insights - Saved Justifications & Review



HRBA Insights screenshot showing saved justifications, aggregated views, and raw saved entries.

Visual Focus

HRBA Insights screenshot showing saved justifications, aggregated views, and raw saved entries.

The HRBA Insights page is the review workspace for previously saved AAAQ analyses.

What You See Here:

- **Saved Justifications table** - A structured view of all persisted HRBA justifications
- **Filters and grouping** - Lets you narrow results by document or analysis context
- **Raw entries** - Full stored payloads for auditing, export, or manual inspection

What It Represents:

- Each row corresponds to a saved analysis record previously stored from the HRBA match workflow
- This page is not generating new analyses; it is reading historical records from storage

When to Use:

- To review previously generated human-rights-based analyses
- To compare justification quality across documents
- To audit the saved evidence that supports HRBA findings

CHAPTER 13

Developer & Admin - Setup, Env vars, Troubleshooting

Chapter Focus

This section covers environment variables, common admin tasks, and troubleshooting steps. Environment variables (important): Common admin tasks:

Visual chapter opener generated from the current SIPMT interface

Developer & Admin - Setup, Env vars, Troubleshooting

Visual Focus

Developer and admin screenshot with environment, maintenance, and troubleshooting utilities.

This section covers environment variables, common admin tasks, and troubleshooting steps.

Environment variables (important):

- DATABASE_URL: full PostgreSQL connection string. If not present, DATABASE_PATH (SQLite) is used.
- DATABASE_PATH: path to fallback SQLite database (default ./data/sipmt.db).
- FORCE_POSTGRES: if set to 1, force use of DATABASE_URL even when it points to localhost.
- SEMANTIC_LLM_PROVIDER: gemini or ollama.
- GEMINI_API_KEY: required when SEMANTIC_LLM_PROVIDER=gemini.
- OLLAMA_BASE_URL: local Ollama endpoint (default http://localhost:11434).
- OLLAMA_MODEL: Ollama model name (e.g., qwen2.5:14b-instruct).
- OLLAMA_TIMEOUT: HTTP timeout for Ollama requests (seconds). Default 120.
- ENABLE_GEOCODING: true/false to enable geocoding of extracted locations.

Common admin tasks:

- Initialize DB and directories:

```
bash
```

```
python setup.py
```

- Migrate existing SQLite demo data to Postgres:

```
bash
```

```
python scripts/migrate_sqlite_to_postgres.py --sqlite-path ./data/sipmt.db --postgres-url <YOUR_URL>
```

- Pull Ollama model (local machine):

```
bash
```

```
ollama pull qwen2.5:14b-instruct
```

Troubleshooting tips:

- App fails to start: run bash scripts/pre_deploy_check.sh to locate syntax or config issues.
- Ollama unreachable: confirm OLLAMA_BASE_URL, try curl http://localhost:11434/.
- Slow semantic analysis: increase OLLAMA_TIMEOUT, warm the model by making a small request, or choose a smaller model.
- Missing extracted text: verify OCR settings and re-run the document extractor pipeline.

Logs and diagnostics:

- Check logs/ for recent app logs.
- Streamlit console shows startup errors; consult server logs for stack traces.

Security & secrets:

- Never commit secrets (API keys, DB passwords) to git. Use platform secrets or an .env file excluded from VCS.

- Rotate API keys if they are accidentally exposed.

CHAPTER 14

Tips & Best Practices

Chapter Focus

Operational guidance and visual reference for this SIPMT application section.

Visual chapter opener generated from the current SIPMT interface

Tips & Best Practices

Document Upload Tips

- **Quality matters**: Ensure scanned PDFs have good resolution (200 DPI) for better text extraction
- **File naming**: Use descriptive names (e.g., Policy_2024_Social_Inclusion.pdf) for easy tracking
- **Organize by type**: Categorize documents correctly - this helps template matching
- **Version control**: Upload versioned documents (v1, v2) to track evolution

Template Design

- **Be specific**: More detailed field descriptions help the matcher find values
- **Use consistent naming**: Similar fields across templates should have similar names
- **Set realistic priorities**: Mark truly required fields, keep others optional
- **Test first**: Create a template and test with 1-2 documents before batch processing

Report Generation

- **Review low-confidence fields**: Always verify extractions with <70% confidence
- **Document sources**: Keep records of which documents were used for each report
- **Export regularly**: Save reports as CSV/JSON for backup and integration with other tools

Compliance Analysis

- **Read recommendations**: AI-generated gaps often contain actionable improvements
- **Cross-check references**: Verify external links in recommendations are correct
- **Track changes**: Re-run analysis after document updates to show progress
- **Use version history**: Review how reference templates evolve over time

Performance Tips

- **Batch processing**: Upload multiple documents at once for efficiency
- **Local Ollama**: Use Ollama instead of Gemini for faster, offline analysis
- **Disable geocoding**: Turn off if you don't need location mapping (faster processing)
- **Clean up old data**: Archive or delete documents you no longer analyze

Troubleshooting

- **Upload fails**: Ensure file is <20 MB and is PDF/DOCX/XLSX format
- **No extracted text**: Use "Re-process" button on Documents page
- **Compliance analysis slow**: Check internet connection (for Gemini) or Ollama service status
- **Map shows no locations**: Ensure documents were categorized as "Policies"

Support, Versioning, and Deployment Notes

Developer Contact

Opus Labs d.o.o. Novi Sad
Nikolajevska 2
21000 Novi Sad, Serbia
www.opus.rs
office@opus.rs

Versioning

Document: SIPMT Playbook

Edition: March 2026

Source: Generated from the current in-application Help page and captured screenshots

Deployment Notes

- Primary local workflow uses PostgreSQL when DATABASE_URL is configured; SQLite remains available as the local/demo fallback.
- Refresh the SQLite demo mirror with `scripts/sync_postgres_to_sqlite.py` after direct PostgreSQL-side changes.
- Run `scripts/pre_deploy_check.sh` before releases so syntax checks and mirror refresh complete consistently.
- Streamlit Cloud deployments rely on `.streamlit/config.toml` for headless startup behavior.

For deployment, configuration, or support questions, contact the developer using the details above.